THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 1: **Patterns in Mathematics** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which triangular number comes just after 276? 1

(a) 300 **(b)** 301 **(c)** 299 **(d)** 323

2. A pattern of stacked sticks uses 3 sticks for 1 triangle and 2 more sticks for each extra triangle added in a row. Sticks for 15 triangles: 1

(a) 31 **(b)** 33 **(c)** 30 **(d)** 32

3. The sum of the first 29 natural numbers added to the sum of the first 30 natural numbers is: 1

(a) 900 **(b)** 901 **(c)** 899 **(d)** 929

4. Powers of 2: $2, 4, 8, 16, \ldots$ The 8th term is: 1

(a) 384 **(b)** 254 **(c)** 256 **(d)** 258

5. The nth term of the pattern $13, 21, 29, 37, \ldots$ is $a + (n-1)d$. The 11th term is: 1

(a) 85 **(b)** 94 **(c)** 93 **(d)** 101

6. In the pattern $2, 11, 20, 29, \ldots$ which term number equals 110? 1

(a) 14 **(b)** 13 **(c)** 15 **(d)** 12

7. What is the sum of the first 18 even numbers $2 + 4 + 6 + \ldots$? 1

(a) 324 **(b)** 360 **(c)** 342 **(d)** 343

8. Which triangular number comes just after 378? 1

(a) 405 **(b)** 407 **(c)** 406 **(d)** 433

9. A pattern of stacked sticks uses 3 sticks for 1 triangle and 2 more sticks for each extra triangle added in a row. Sticks for 20 triangles: 1

(a) 40 **(b)** 41 **(c)** 42 **(d)** 43

10. The sum of the first 34 natural numbers added to the sum of the first 35 natural numbers is: 1

(a) 1259 **(b)** 1224 **(c)** 1226 **(d)** 1225

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Is 62 a term of $6, 11, 16, 21, \ldots$? Give a reason. 2

12. Find the sum of the first 35 odd numbers and the first 35 even numbers. 2

13. Find the nth term formula of $11, 15, 19, 23, \ldots$ and the 18th term. 2

14. Is 81 a term of $2, 8, 14, 20, \ldots$? Give a reason. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. How many triangles' worth of sticks: a row of 18 triangles needs 3 sticks for the first triangle and 2 more for each new one. Find the sticks needed and the number of triangles that can be made with 47 sticks. 3

16. Find the sum of all odd numbers from 1 to 89. Hence find the sum of all even numbers from 2 to 90. 3

17. Write the 28th and 29th triangular numbers and check that they add to a square. 3

OR

How many triangles' worth of sticks: a row of 21 triangles needs 3 sticks for the first triangle and 2 more for each new one. Find the sticks needed and the number of triangles that can be made with 53 sticks. 3

18. Find the sum of all odd numbers from 1 to 99. Hence find the sum of all even numbers from 2 to 100. 3

OR

Write the 32nd and 33rd triangular numbers and check that they add to a square. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A pattern of squares: the first figure has 8 squares and each next figure has 6 more. Find the number of squares in the 9th figure, and the first figure that has more than 106 squares. 5

OR

Show that the sum of the cubes of the first 11 numbers is a perfect square, and find the sum of the first 11 numbers. 5

20. A pattern of squares: the first figure has 4 squares and each next figure has 7 more. Find the number of squares in the 11th figure, and the first figure that has more than 124 squares. 5

OR

Show that the sum of the cubes of the first 12 numbers is a perfect square, and find the sum of the first 12 numbers. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND